

# Lab 1: Flutter Setup + Hello World

## Structure

```
lab1_hello_world/  
└─ lib/  
    └─ main.dart
```

## main.dart

```
import 'package:flutter/material.dart';  
  
void main() {  
  runApp(MyApp());  
}  
  
class MyApp extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: Scaffold(  
        appBar: AppBar(title: Text("Hello App")),  
        body: Center(  
          child: Text("Hello World"),  
        ),  
      ),  
    );  
  };  
}
```

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# Lab 2: Dart Basics (Variables & Functions)

## Structure

```
lab2_dart_basics/  
└─ lib/  
    └─ main.dart
```

## **main.dart**

```
void main() {  
  int a = 10;  
  int b = 20;  
  
  print("Sum: ${add(a, b)}");  
}  
  
int add(int x, int y) {  
  return x + y;  
}
```

---

# Lab 3: Simple UI Text App

## Structure

```
lab3_simple_ui/  
└── lib/  
    └── main.dart
```

## main.dart

```
import 'package:flutter/material.dart';  
  
void main() => runApp(MyApp());  
  
class MyApp extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: Scaffold(  
        body: Center(  
          child: Text(  
            "Welcome Student",  
            style: TextStyle(fontSize: 24),  
          ),  
        ),  
      ),  
    );  
  }  
}
```

---

# Lab 4: ListView Dynamic List

## Structure

```
lab4_listview/  
└── lib/  
    └── main.dart
```

## main.dart

```
import 'package:flutter/material.dart';  
  
void main() => runApp(MyApp());  
  
class MyApp extends StatelessWidget {  
  final List<String> items = ["Apple", "Banana", "Mango", "Orange"];  
  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: Scaffold(  
        appBar: AppBar(title: Text("ListView")),  
        body: ListView.builder(  
          itemCount: items.length,  
          itemBuilder: (context, index) {  
            return ListTile(title: Text(items[index]));  
          },  
        ),  
      ),  
    );  
  }  
}
```

---

# Lab 5: Navigation with Arguments

## Structure

```
lab5_navigation/  
└── lib/  
    ├── main.dart  
    └── second.dart
```

## main.dart

```
import 'package:flutter/material.dart';  
import 'second.dart';  
  
void main() => runApp(MyApp());  
  
class MyApp extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(home: FirstScreen());  
  }  
}  
  
class FirstScreen extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return Scaffold(  
      appBar: AppBar(title: Text("First Screen")),  
      body: Center(  
        child: ElevatedButton(  
          onPressed: () {  
            Navigator.push(  
              context,  
              MaterialPageRoute(  
                builder: (context) => SecondScreen(data: "Hello from First"),  
              ),  
            );  
          },  
          child: Text("Go"),  
        ),  
      ),  
    );  
  }  
}
```

## **second.dart**

```
import 'package:flutter/material.dart';

class SecondScreen extends StatelessWidget {
  final String data;

  SecondScreen({required this.data});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Second Screen")),
      body: Center(child: Text(data)),
    );
  }
}
```

---

# Lab 6: Responsive Layout (Flex, Expanded)

## Structure

```
lab6_responsive/  
├── lib/  
│   └── main.dart
```

## main.dart

```
import 'package:flutter/material.dart';  
  
void main() => runApp(MyApp());  
  
class MyApp extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: Scaffold(  
        body: Row(  
          children: [  
            Expanded(child: Container(color: Colors.red)),  
            Expanded(child: Container(color: Colors.green)),  
            Expanded(child: Container(color: Colors.blue)),  
          ],  
        ),  
      ),  
    );  
  }  
}
```

---

# Lab 7: TextField Input

## Structure

```
lab7_input/  
└─ lib/  
    └─ main.dart
```

## main.dart

```
import 'package:flutter/material.dart';  
  
void main() => runApp(MyApp());  
  
class MyApp extends StatefulWidget {  
  @override  
  _MyAppState createState() => _MyAppState();  
}  
  
class _MyAppState extends State<MyApp> {  
  String text = "";  
  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: Scaffold(  
        body: Column(  
          children: [  
            TextField(  
              onChanged: (value) {  
                setState(() {  
                  text = value;  
                });  
              },  
            ),  
            Text(text),  
          ],  
        ),  
      ),  
    );  
  }  
}
```

---

# Lab 8: Firebase Authentication

## Add dependency

```
firebase_auth  
firebase_core
```

## main.dart

```
import 'package:flutter/material.dart';  
import 'package:firebase_core/firebase_core.dart';  
import 'package:firebase_auth/firebase_auth.dart';  
  
void main() async {  
  WidgetsFlutterBinding.ensureInitialized();  
  await Firebase.initializeApp();  
  runApp(MyApp());  
}  
  
class MyApp extends StatelessWidget {  
  final auth = FirebaseAuth.instance;  
  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: Scaffold(  
        body: Center(  
          child: ElevatedButton(  
            onPressed: () async {  
              await auth.createUserWithEmailAndPassword(  
                email: "test@test.com",  
                password: "123456",  
              );  
            },  
          ),  
          child: Text("Register"),  
        ),  
      ),  
    ),  
  );  
}  
}
```

---

# Lab 9: Form Validation

## Structure

```
lab9_form/  
└── lib/  
    └── main.dart
```

## main.dart

```
import 'package:flutter/material.dart';  
  
void main() => runApp(MyApp());  
  
class MyApp extends StatelessWidget {  
  final _formKey = GlobalKey<FormState>();  
  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: Scaffold(  
        body: Form(  
          key: _formKey,  
          child: Column(  
            children: [  
              TextFormField(  
                validator: (value) {  
                  if (value!.isEmpty) return "Enter name";  
                  return null;  
                },  
            ),  
              ElevatedButton(  
                onPressed: () {  
                  if (_formKey.currentState!.validate()) {  
                    print("Valid");  
                  }  
                },  
                child: Text("Submit"),  
              ),  
            ],  
          ),  
        ),  
      ),  
    );  
  }  
}
```

---

# Lab 10: API Fetch (http)

 **Add dependency(in pubspec.yaml)**

http

 **main.dart**

```
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  Future fetchData() async {
    final res = await http.get(Uri.parse("https://jsonplaceholder.typicode.com/posts"));
    return jsonDecode(res.body);
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        body: FutureBuilder(
          future: fetchData(),
          builder: (context, snapshot) {
            if (snapshot.hasData) {
              return ListView.builder(
                itemCount: snapshot.data.length,
                itemBuilder: (context, index) {
                  return Text(snapshot.data[index]['title']);
                },
              );
            }
            return CircularProgressIndicator();
          },
        ),
      ),
    );
  }
}
```

---

## Lab 11: Location (Geolocator)

## Add dependency(in pubspec.yaml)

geolocator

## main.dart

```
import 'package:flutter/material.dart';
import 'package:geolocator/geolocator.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  Future<Position> getLocation() async {
    return await Geolocator.getCurrentPosition();
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        body: FutureBuilder(
          future: getLocation(),
          builder: (context, snapshot) {
            if (snapshot.hasData) {
              return Text(
                "Lat: ${snapshot.data!.latitude}, Lng: ${snapshot.data!.longitude}",
              );
            }
            return CircularProgressIndicator();
          },
        ),
      ),
    );
  }
}
```